2 How to Use

2.1 To Select a Machining Level

High accuracy mode A is equipped with a function to improve shape accuracy and another to improve surface quality. There are 6 machining levels available in the default settings, such as one that focuses on shape accuracy and another that focuses on surface quality. Select one from these six machining levels depending on the machining content. (Note, in the default settings, the function that improves the surface quality uses smooth override.)
In addition, there are two more levels other than the above levels that can be defined by users so that the machining levels can be adjusted to the optimum. For detail, refer to the “4 Detailed explanations and adjustments of parameter”.

Machining level default setting

Machining level	Purpose (Emphasis)	Contents	M code
1	Shape accuracy	The shape accuracy is emphasized. (First recommended for emphasizing shape accuracy) The surface quality is equivalent to the level when high accuracy mode is not used. Machining time becomes a little longer because it decelerates in accordance with a shape.	M260
2	Shape accuracy (High)	The shape accuracy is more emphasized than the machining level 1. The surface quality is less than when high accuracy mode is not used. Machining time becomes a little longer because it decelerates in accordance with a shape. In addition, the top speed of a command is limited. (F5000 is the default value.)	M261
3	Shape accuracy + face dignity	Both the shape accuracy and face dignity are emphasized. As the deceleration time become longer, the machining time may become very long.	M262
4	Face dignity	The face dignity is emphasized (first recommended for emphasizing face dignity). Use this if smooth face dignity is necessary. However, shape accuracy becomes bad.	M263
5	Face dignity (High)	The face dignity is emphasized more than the machining level 4. Use this if smoother face dignity is necessary. However, shape accuracy becomes worse.	M264
6	High speed	High speed is emphasized. The shape accuracy is a little higher than when high accuracy mode is not used. The face dignity is in the same level. If you are not sure of this shape accuracy emphasis or the face dignity emphasis, it is recommended to use this machining level.	M265
7/8	User definition	If it is necessary to set other than the above six, set a value of the user parameter referring to the parameter values of other levels.	M266 to M267

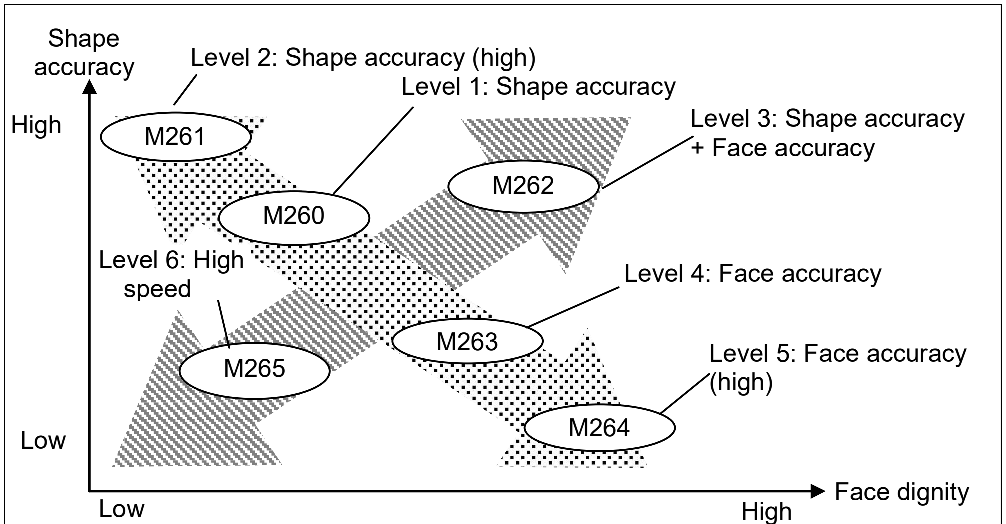


Image of machining levels

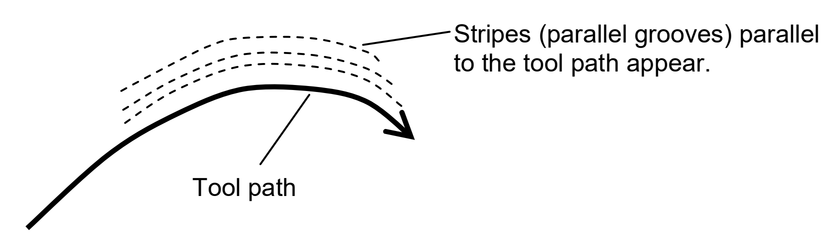
Points for selection of machining level

1. Select the shape accuracy emphasis or the face dignity emphasis
   Decide which of the shape accuracy or the face dignity is to be emphasized. If the shape accuracy is emphasized, the tool path is faithfully controlled by the command path. If the face dignity is emphasized, it is controlled to improve smoothness of the machining face.

   - Shape accuracy emphasis
     → Select the machining level 1 (shape accuracy, M260).
   - Face dignity emphasis
     → Select the machining level 4 (face dignity, M263).
   - If you are not sure which should be emphasized
     → Select the machining level 1 (shape accuracy, M260) or machining level 6 (high-speed, M265).

2. Decide which of the shape accuracy or face dignity is necessary for the machining.
   Depending on the stripes (parallel grooves) appering on the machining face, you may know which of the shape accuracy or the face dignity is necessary to be emphasized.

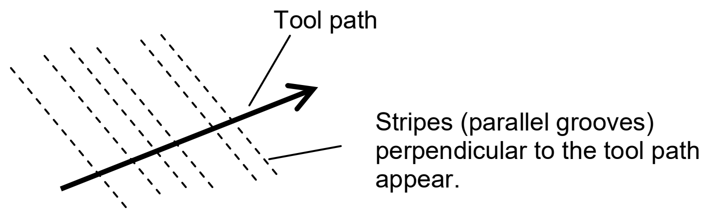
(1) Stripes (parallel grooves) parallel in the tool ongoing direction appear.



Presumed cause
There ia a possibility that the stripes (parallel grooves) may appear due to a shape error. Especially they tend to appear if the commanded speed is too fast or the curvature is too small.

Recommend machining level
Machining level 1 (shape accuracy, M260)

(2) Stripes (parallel grooves) perpendicular to the tool path in the ongoing direction appear.



Presumed cause
The minute linear command created by CAM or the like for a curved shape is polygonal, so it tends to generate stripes (parallel grooves). Especially they tend to appear if the speed is too slow or the curvature is too large. They may be none by minimizing the tolerance of CAM.

Recommend machining level
Machining level 4 (face dignity, M263)

(3) If you want to emphasize the shape accuracy or the face dignity more
Use the machining level 2 (shape accuracy (High), M261) or the machining level 5 (face dignity (High), M264). However, the machining level 2 may take more machining time than the machining level 1. The machining level 5 worsens the shape accuracy more than the machining level 4.

(4) The machining level 3 (shape accuracy + face dignity, M262) may take very long machining time.